

The Logic of Logical Atomism

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1 Logical Atomism

Wittgenstein (1921) and Russell (1956 [1918], 1924/1956) defend versions of *Logical Atomism*. These views suggest:

(LA) Every proposition is a truth-functional combination of elementary propositions.

Most existing formal discussions are couched in terms of Boolean algebras (see, e.g., Skyrms (1993)). LA and related principles can also be formulated in the object language of higher-order logic.

2 Boolean Algebras

Assume for now that propositions form a Boolean algebra $\mathfrak{A}$.

Definition 1. A Boolean Algebra is a structure $\mathfrak{A} = \langle A, \neg, \wedge, \vee \rangle$ such that $\varphi = \psi$ whenever $\varphi \leftrightarrow \psi$ is a tautology.

Standard definitions and results are taken from Koppelberg (1989) and Givant and Halmos (2009). We also need:

Definition 2. For $E \subseteq B$:

- $D \triangleleft_\omega E$ if $D = T \cup \{-f : f \in F\}$ for sets $T, F \subseteq_\omega E$ such that $T \cap F = \emptyset$.
- $D \triangleleft E$ if $D = T \cup \{-f : f \in F\}$ for sets $T, F \subseteq E$ such that $T \cap F = \emptyset$ and $T \cup F = E$.

2.1 Free Boolean Algebras

Bell and Demopoulos (1996) suggest capturing LA as follows:

Definition 3. $\mathfrak{A}$ is independently generated if for some $E \subseteq A$:

- E is independent, i.e., $\bigwedge D > \perp$ for every $D \triangleleft_\omega E$.
- E generates $\mathfrak{A}$, i.e., no proper subalgebra of $\mathfrak{A}$ includes E .

Independent generation is equivalent to being free:

Definition 4.

- $\mathfrak{A}$ is free relative to a class of BAs $\mathcal{C}$ if for some $E \subseteq A$, for every $\mathfrak{B}$ in $\mathcal{C}$ and function $f : E \rightarrow B$, there is a unique homomorphism $\hat{f}$ from $\mathfrak{A}$ to $\mathfrak{B}$ extending f .
- $\mathfrak{A}$ is free if $\mathfrak{A}$ is free relative to the class of all BAs.

This makes sense if there are all these other algebras. If this is just a mathematical model, we might opt for a variant, such as:

Definition 5.

- $\mathfrak{A}$ is weakly internally free if $\mathfrak{A}$ is free relative to $\{\mathfrak{A}\}$.
- $\mathfrak{A}$ is strongly internally free if $\mathfrak{A}$ is free relative to the class of subalgebras of $\mathfrak{A}$.

Even more restrictively, you might consider just the truth values:

Definition 6.

- $\mathfrak{A}$ is truth-value free if $\mathfrak{A}$ is free relative to $\{\mathbf{2}\}$.

This captures the idea that the truth values of all propositions are uniquely determined by the truth values of the elementary ones.

Pleasingly, all these conditions are equivalent.

2.2 Double Powerset Algebras

Suszko (1968) and Moss (2012) effectively suggest capturing LA by requiring $\mathfrak{A}$ to be a double powerset algebra $\mathcal{PP}(E)$.

If $\mathfrak{A}$ is finite, this is equivalent to the previous conditions. But if $\mathfrak{A}$ is infinite, being free entails being atomless.

According to LA, quantifiers express conjunctions and disjunctions, which are plausibly glbs/lubs of instances. What happens if we take these into account? Assume from now on that $\mathfrak{A}$ is complete.

2.3 Complete Boolean Algebras

Definition 7.

- $\mathfrak{A}$ is completely independently generated (CIG) if for some $E \subseteq A$:
 - E is completely independent, i.e., $\bigwedge D > \perp$ for every $D \triangleleft E$.
 - E completely generates $\mathfrak{A}$, i.e., no proper complete subalgebra of $\mathfrak{A}$ includes E .
- $\mathfrak{A}$ is completely free relative to a class of CBAs $\mathcal{C}$ if for some $E \subseteq A$, for every $\mathfrak{B}$ in $\mathcal{C}$ and function $f : E \rightarrow B$, there is a unique complete homomorphism $\hat{f}$ from $\mathfrak{A}$ to $\mathfrak{B}$ extending f .
- $\mathfrak{A}$ is completely free (CF) if $\mathfrak{A}$ is free relative to the class of all CBAs.
- $\mathfrak{A}$ is completely weakly internally free (CWIF) if $\mathfrak{A}$ is completely free relative to $\{\mathfrak{A}\}$.
- $\mathfrak{A}$ is completely strongly internally free (CSIF) if $\mathfrak{A}$ is completely free relative to the class of subalgebras of $\mathfrak{A}$.
- $\mathfrak{A}$ is completely truth-value free (CTVF) if $\mathfrak{A}$ is completely free relative to $\{\mathbf{2}\}$.

Since $\mathfrak{A}$ is complete, we may assume (up to isomorphism) that $\mathfrak{A} = \mathfrak{B} \times \mathfrak{C}$ where $\mathfrak{B}$ is an atomless CBA and $\mathfrak{C}$ is an atomic CBA.

We have a good understanding of some of these conditions:

Proposition 8. $\mathfrak{A}$ is completely truth-value free iff $\mathfrak{C}$ is a double powerset algebra.

This is equivalent to a version of the idea that elementary propositions determine possible worlds:

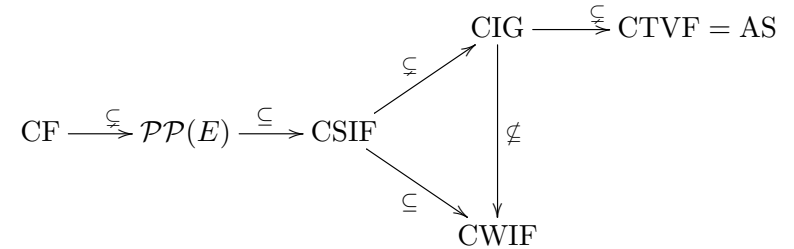
Definition 9. $\mathfrak{A}$ is alethically supervenient (AS) if for some $E \subseteq A$, for every $D \triangleleft E$, $\bigwedge D$ is an atom.

Proposition 10. $\mathfrak{A}$ is completely truth-value free iff $\mathfrak{A}$ is alethically supervenient.

Proposition 11. $\mathfrak{A}$ is completely independently generated iff $\mathfrak{C} \cong \mathcal{PP}(E)$ for some set E and $\mathfrak{B}$ is completely generated by a set of size $\leq |E|$.

Proposition 12 (based on Gaifman (1964), Hales (1964)). $\mathfrak{A}$ is completely free iff $\mathfrak{A}$ is a finite double powerset algebra.

The internal notions are much more difficult to understand. Our main results are as follows:



We can also show $\text{CWIF} \not\subseteq \text{CIG}$ (and so $\text{CWIF} \not\subseteq \text{CSIF}$) if:

Conjecture 13. There is an infinite CBA without non-trivial complete endomorphisms.

3 Logical Atomism in Higher-Order Logic

Russell and Wittgenstein did not believe in complex propositions. A sentence like $A \vee B$ is not made true by a single disjunctive fact, but by multiple atomic facts (e.g. Russell (1956 [1918]) pp. 38–39 in Routledge edition).

- Truth functionally equivalent sentences, like $(A \wedge B) \vee \neg C$ and $(A \vee \neg C) \wedge (B \vee \neg C)$, are made true by the same combination of facts. This motivates *Booleanism*, by which we will mean the intersubstitutivity (with capture) of truth-functional equivalents in all contexts.¹
- How should we interpret existential quantification into sentence position? We will adopt a reading of these quantifiers in which existential generalization into the position of a complex proposition (e.g. inferring $\exists tp.Fp$ from $F(A \vee B)$), is legitimate. Indeed, according to the Tractarian account of existential quantification an existential is really an infinite disjunction of instances of a propositional function. So this existential can be unproblematically true, in virtue of the the elementary truths, in the same way a disjunction can be.² This seems to push us all the way to Classicism, plus an axiom saying that existentials and universals are equivalent to disjunctions and conjunctions.
- Russell, by contrast, does not think that a universal is made true by the totality of its instances – you need a further ‘that’s all’ fact. But he seems to posit such a fact for each (ramified) type, allowing, again, for an interpretation where existential generalization into the position of complex terms is valid.³

¹‘With capture’ means that I can substitute $p \wedge q$ with $q \wedge p$ in $\lambda pq.(p \wedge q)$ to get $\lambda pq.(q \wedge p)$, for instance.

²In what sense, then, do complex propositions not exist? It is unclear what the Tractarian view is here, since it does not treat higher-order quantification.

³Klement (forthcoming) worries about there being too many ‘that’s all’ facts, one for each generalization. However on Klement’s substitutional interpretation of

Framework:

Classicism with $\Box$ Boolean Completeness.

A non-logical constant, $E : t \rightarrow t$, representing the elementary propositions, and an axiom stating that it is rigid.

Optionally, we could add an axiom corresponding to the Tractarian idea that quantificational claims are conjunctions or disjunctions.

Tractarian Quantification $\Box \forall tp(\forall \sigma y(p \leq_\sigma Fy) \rightarrow p \leq_t \forall \sigma F)$

4 Some informal atomist ideas

The Truth-Table Postulate It’s a tautology that

- Any assignment of truth-values to the elementary propositions can be extended to a consistent assignment of truth-values to all propositions.
- The truth values of all propositions are uniquely determined by the assignment of truth values to the elementary propositions (i.e. this extension is unique).

Alethic Supervenience. It’s a tautology that, for any division of elementary propositions into two classes, the proposition stating that the first class consists of truths and the second class consists of falsehoods is

- Possible.
- Settles every question.

Generation It’s a tautology that the smallest collection of propositions containing the elementary propositions and closed under negation and arbitrary conjunctions includes all propositions.

quantification over propositional functions in Russell it’s not obviously a problem, since a ‘that’s all’ fact for the individuals and simple relations serves as a ‘that’s all’ fact for the predicates definable (with parameters) from those individuals and simple relations.

5 Definitions

Definition 14 (Tautologousness). *We'll say that p is tautologous (or necessary), written $\Box p$, iff $p = \top$, where $\top$ is some fixed tautology. p entails q means $\Box(p \rightarrow q)$.*

Definition 15 (Truth assignment). *A truth-value assignment is a binary relation $V_0 : t \rightarrow t \rightarrow t$ that is (i) functional, (ii) its domain are all and only the elementary propositions, (iii) only $\top$ and $\perp$ are in its range.*

Definition 16 (Valuation). *A valuation is a binary relation $V : t \rightarrow t \rightarrow t$ that is (i) functional, (ii) its domain includes every proposition, (iii) only $\top$ and $\perp$ are in its range and (iv) it preserves GLBs and negation, in the sense that for any $X : t \rightarrow t$, $Vp\top$ for all p such that Xp , iff $V(\text{GLB}(X))\top$ where $\text{GLB}(X)$ is the GLB of the X s, and Vpq iff $V\neg p\neg q$.*

We say a valuation V extends a truth-value assignment V_0 iff for every elementary proposition p , and every proposition q , V_0pq iff Vpq .

Definition 17 (The world according to V). *Let V be a valuation (or truth-value assignment): V tells us, in a consistent manner, what the truth value of every (elementary) proposition is. The world according to V , or W_V , is the claim that the propositions that V says are true are true, and the propositions V says are false are false. That is, W_V is the GLB of propositions p that V relates to $\top$ and propositions $\neg p$ where V relates p to $\perp$.*

Proposition 18. *For any valuation V , the world according to V , W_V , is a world proposition.*

Note that every world proposition W determines a unique valuation $V[p] = \top$ if $\Box(W \rightarrow p)$ and $V[p] = \perp$ if $\Box(W \rightarrow \neg p)$. So world propositions are in one-to-one correspondence with valuations.

Definition 19. *An operator X is closed iff it applies to all the elementary propositions, applies to $\neg p$ whenever it applies to p and applies to $\text{GLB } Y$ whenever it applies to every Y proposition.*

The Truth-Table Postulate Necessarily, every truth-value assignment V_0 extends to a unique valuation V up to extension:

$$\Box \forall_{t \rightarrow t} V_0(\text{TA } V_0 \rightarrow \exists_{t \rightarrow t} V(\text{Val } V \wedge \forall pq(V_0pq \rightarrow Vpq) \wedge \forall_{t \rightarrow t} U(\text{Val } U \wedge \forall_t pq(V_0pq \rightarrow Upq) \rightarrow \forall pq(Vpq \leftrightarrow Upq))).^4$$

Alethic Supervenience Necessarily, for every truth-value assignment V_0 , W_{V_0} is a world proposition. $\forall_{t \rightarrow t} V_0(\text{TA } V_0 \rightarrow \text{World } W_{V_0})$.

Generation Every closed property of propositions contains all propositions: $\forall_{t \rightarrow t} X(\text{Closed } X \rightarrow \forall_t p.Xp)$

6 The Truth Table Postulate and Alethic Supervenience

Proposition 20. *Assuming Tractarian Quantification and $\Box$ Rigid Comprehension, the Truth-Table Postulate entails Alethic Supervenience.*

Proposition 21. *Alethic Supervenience entails the Truth Table Postulate.*

Theorem 22. *In the presence of $\Box$ Rigid Comprehension and Tractarian Quantification, Alethic Supervenience and the Truth-Table Postulate are equivalent.*

Rigid Comprehension is not obviously motivated from a logical atomist perspective. Without it, the implication from Alethic Supervenience to the Truth-Table Postulate still holds, but we do not know about the converse implication. In this setting Alethic Supervenience may be stronger, in contrast to the situation with Boolean algebras (cf proposition 10).

⁴The necessarily in front of this postulate is needed for some of the results; one might try to motivated it from the idea that the statements of the Tractatus are say nothing about the world/are tautologous.

7 Generation?

We have explored two ways of articulating the idea that all truths boil down to the truth values of the elementary propositions: the truth-table postulate and alethic supervenience. These principles tell us, in different ways, that all propositions are settled by the elementary proposition. Do these principles imply that all propositions are truth-functional combinations of the elementary propositions: does every proposition fall under every property applying to the elementary propositions, closed under GLBs and negation?

Proposition 23. *There is a model of Classicism in which the following hold:*

1. *E is rigid*
2. *S5 (and thus also Tractarian Quantification).*
3. *Alethic Supervenience*
4. *The Truth-Table Postulate*
5. *$\neg$ Generation*
6. *$\Diamond\neg$ Rigid Comprehension^t.*

In this model the propositions are modeled by a complete truth-value free cBa that is not completely independently generated (the product of a double powerset and a complete atomless Boolean algebra).

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